

**FACULTY OF MECHANICAL ENGINEERING
CAN THO UNIVERSITY OF TECHNOLOGY**



PHÁT TRIỂN ỨNG DỤNG IOT *(IoT application development)*

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- ❖ **Môn học tham gia giảng dạy:**
 - Nhập môn công nghệ kỹ thuật, phát triển ứng dụng IoT, trí tuệ nhân tạo trong điều khiển, hệ thống điều khiển số, toán kỹ thuật.

Thông tin về học phần

- ❑ Tên học phần bằng tiếng Việt: **Phát triển ứng dụng IoT**
- ❑ Tên học phần bằng tiếng Anh: **IoT Application Development**
- ❑ Mã học phần: **TT188**
- ❑ Số tín chỉ: **03** tín chỉ (**30** tiết lý thuyết, **30** tiết thực hành)
- ❑ Mục tiêu cụ thể: *Giới thiệu các kiến thức cơ bản về Internet vạn vật, các yêu cầu, tiêu chuẩn và ứng dụng của IoT từ đó sinh viên có thiết kế đánh giá các hệ thống theo dõi, điều khiển qua Internet.*
- ❑ Kỹ năng: *hướng dẫn thiết kế các hệ thống theo dõi và điều khiển qua Internet trên hệ thống thực.*

Tài liệu tham khảo

- ❖ Lê Trung Quân, Huỳnh Văn Đặng, và Nguyễn Khánh Thuật, “Giáo trình Công Nghệ Internet of Things và Ứng Dụng”, NXB ĐHQGTPHCM, 2021.
- ❖ Ammar Rayes and Samer Salam, “*Internet of Things from Hype to Reality-The Road to Digitization*”, 3rd Edition, Springer, 2022.
- ❖ Nguyen Tang Kha Duy, Nguyen Dinh Tu, Tra Hoang Son, and Luong Hong Duy Khanh, “*Automated Monitoring and Control System for Shrimp Farms based on Embedded System and Wireless Sensor Network*”, 2015 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT), Coimbatore, India, 2015, pp.1-5.
- ❖ Neil Cameron, “*Electronics Projects with the ESP8266 and ESP32*”, 1st Edition, Apress, 2021.

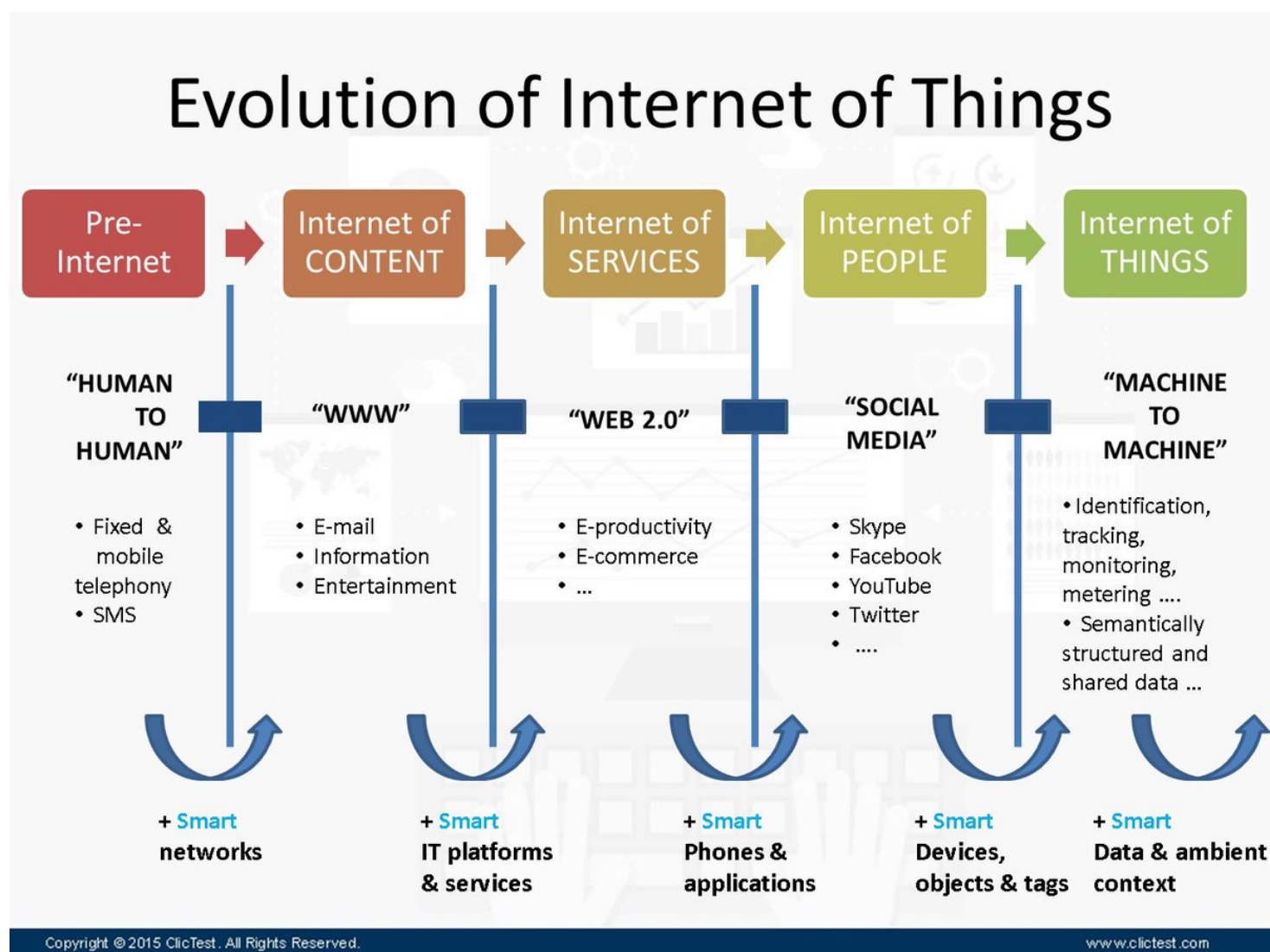
Nội dung

- ❖ Chương 1: Giới thiệu
- ❖ Chương 2: Các lĩnh vực ứng dụng IoT
- ❖ Chương 3: Các kiến trúc mạng và bộ giao thức mạng IoT
- ❖ Chương 4: Các nền tảng phần cứng và phần mềm IoT thông dụng
- ❖ Chương 5: Điện toán đám mây, điện toán cận biên và IoTs
- ❖ Chương 6: Quản lý ngữ cảnh thông minh và trí tuệ nhân tạo trong IoT

CHƯƠNG 1: GIỚI THIỆU

- ❖ **1.1**: Tổng quan về công nghệ Internet vạn vật
- ❖ **1.2**: Xu thế ứng dụng IoTs và các công nghệ liên quan cho đô thị thông minh ở Việt Nam
- ❖ **BÀI TẬP CHƯƠNG 1**

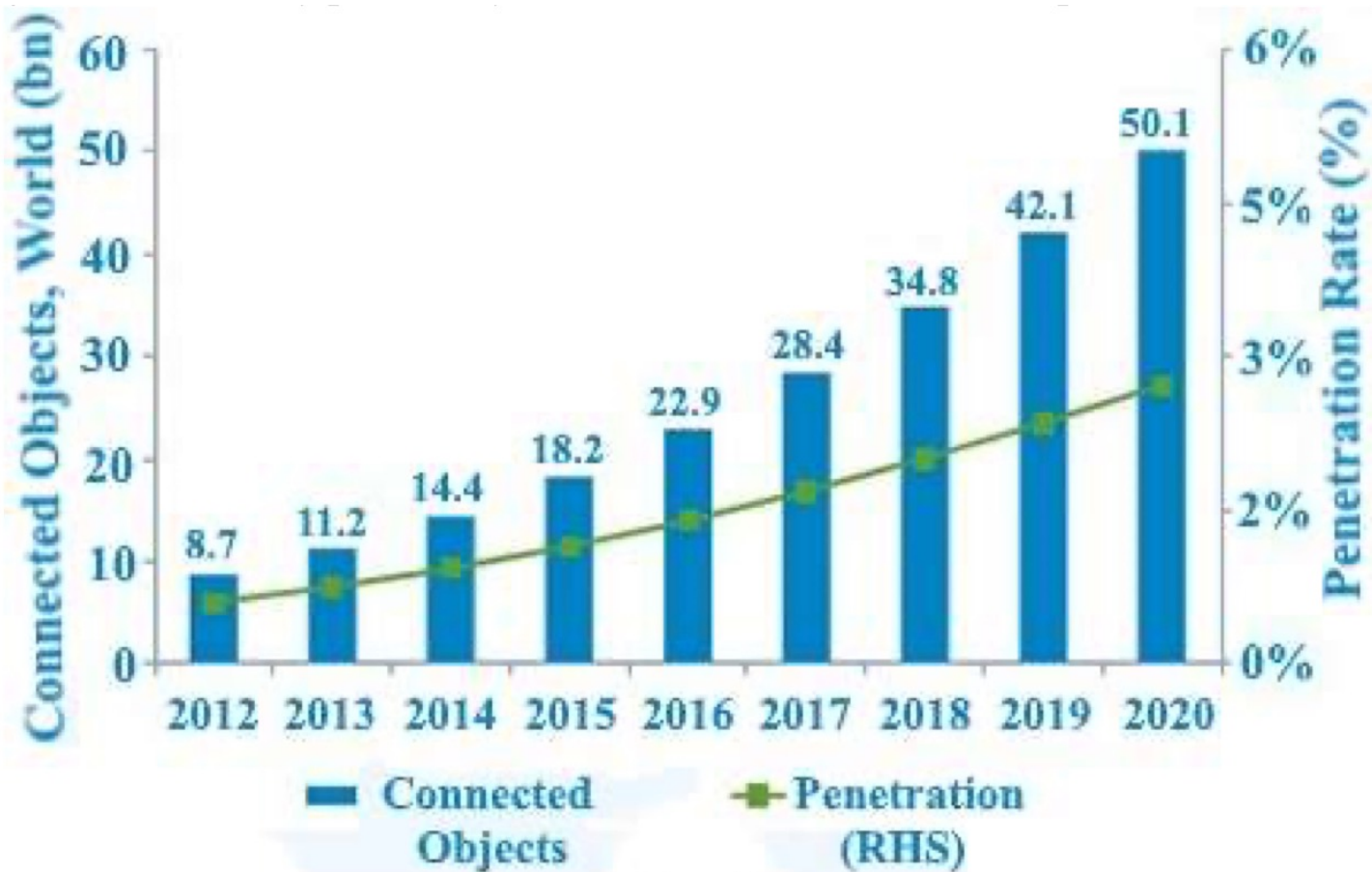
1.1. Tổng quan về công nghệ Internet vạn vật



1.1. Tổng quan về công nghệ Internet vạn vật

- Thế giới hiện có khoảng 20 tỷ thiết bị kết nối Internet và đang tăng trung bình 80 thiết bị mới mỗi giây. Tốc độ tăng này sẽ đạt 50 tỷ thiết bị vào năm 2020.
- Đây là một trong những thống kê của Cisco về viễn cảnh “Internet vạn vật” (Internet of Things – IoTs).
- Công nghệ mạng IoTs hiện đang được phát triển để mọi thiết bị, đồ vật... đều có thể kết nối và tương tác với nhau, giúp chúng ta có thể tra cứu thông tin về thực phẩm ngay trên tủ lạnh, điều khiển ô tô từ xa mà không cần trực tiếp lái xe, hoặc 4 – 5 triệu xe máy ở Hà Nội, hoặc TP. HCM được tích hợp cảm biến để nhà quản lý dễ dàng kiểm soát lưu lượng giao thông.

1.1. Tổng quan về công nghệ Internet vạn vật



Số liệu minh họa số lượng thiết bị IoTs cho đến năm 2020

1.1. Tổng quan về công nghệ Internet vạn vật

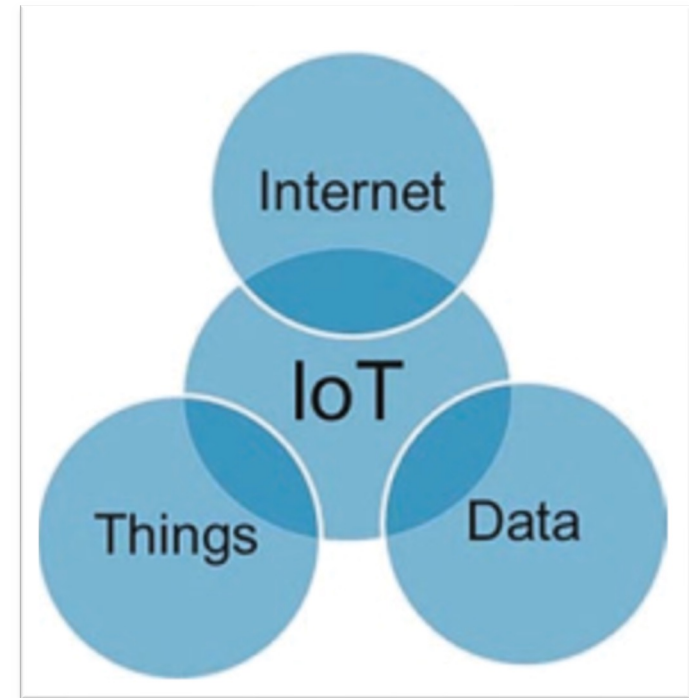
- ❑ Before defining IoT, it may be worthwhile listing the most generic enablement components. In its simple form, IoT may be considered as a network of physical elements empowered by:
 - **Sensors**: to collect information.
 - **Identifiers**: to identify the source of data (e.g., sensors, devices).
 - **Software**: to analyze data.
 - **Internet connectivity**: to communicate and notify.

IoT is the network of things, with clear element identification, embedded with software intelligence, sensors, and ubiquitous connectivity to the Internet.

1.1. Tổng quan về công nghệ Internet vạn vật

Background and More Complete IoT Definition

- ❑ Before we give historical overview of the Internet and consequently delve into the **Internet of Things**, it is worthwhile providing a definition and the fundamental requirements of **IoT** as a basis for the inexperienced reader.
- ❑ We assume that the Internet is well known and bears no further definition. The question is what do we really mean by “**Things**”? Well, things are actually “**anything**” and “**everything**” from appliances to buildings to cars to people to animals to trees to plants, etc.



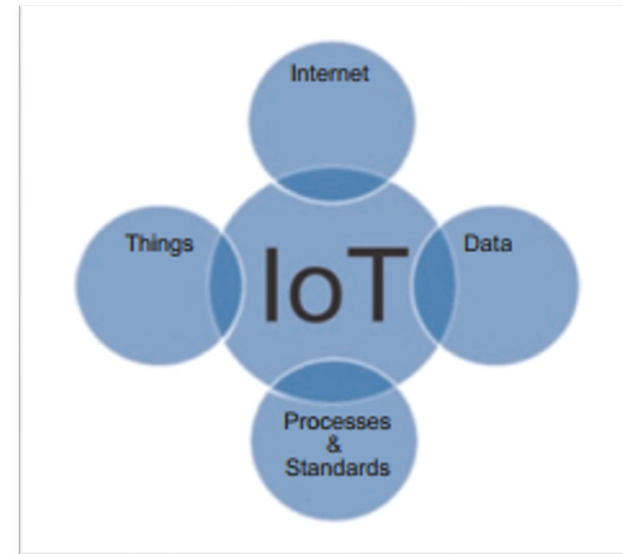
IoT definition in its simplest form

1.1. Tổng quan về công nghệ Internet vạn vật

Background and More Complete IoT Definition

A more complete definition, we believe, should also include “Standards” and “Processes” allowing “Things” to be connected over the “Internet” to exchange “Data” using industry “Standards” that guarantee interoperability and enabling useful and mostly automated “Processes”.

- ❖ **Data:** Converting data into intelligence to make better decisions.
- ❖ **Process:** Delivering the right information to the right person or machine at the right time.
- ❖ **Things:** Physical devices and objects connected to the Internet and each other for intelligent decision-making, often called IoT.

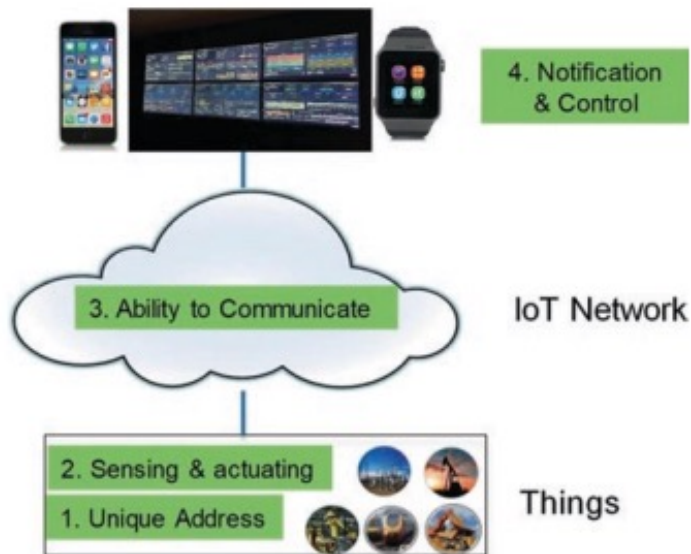


IoT-more complete definition

1.1. Tổng quan về công nghệ Internet vạn vật

How to Monitor and Control Things from Anywhere in the World?

- The **basic requirements** for IoT are the unique identity per “**thing**” (e.g., **IP address**), the ability to communicate between things (e.g., **wireless communications**), and the ability to sense specific information about the thing (**sensors**).



- With these three requirements, one should be able to monitor things from anywhere in the world. Another foundation requirement is a medium to communicate. Such requirement is typically handled by a telecommunications network.

Yêu cầu cơ bản của một giải pháp IoT

1.1. Tổng quan về công nghệ Internet vạn vật

*Why Do We Want to **Monitor** and **Control** Things?*

There are many reasons to monitor and control things remotely over the Internet:

- Monitoring and controlling things by experts (e.g., a patient's temperature or blood pressure while the patient is at the comfort of his or her own home).
- Learning about things by pointing a smartphone to a thing of interest, for instance; searching for things that search engines (e.g., Google) do not provide today (e.g., where are my car keys).
- Allowing authorities to manage things in **smart cities** in an optimal manner (e.g., energy, driver licenses, and other documents from Department Motor Vehicle, senior citizen).
- Providing more affordable entertainment and games for children and adults. All of these are examples of huge business and service opportunities to boost the economic impact for consumers, businesses, governments, hospitals, and many other entities.

1.1. Tổng quan về công nghệ Internet vạn vật

Who will monitor and control?

Generally speaking, **monitoring** and **control** of IoT services may be done by any **person** or any **machine**.

- Homeowner monitoring his own home on a mobile device based on a security system she or he has installed and configured. The homeowner may also control lights, turn on the air conditioning, shut off the heater, etc.
- Another example is for a service provider to monitor and control services for its customers in a **network operations center (NOC)**.



1.1. Tổng quan về công nghệ Internet vạn vật

Who Will Monitor and Control?

Obviously, **security** is a major concern to **prevent** access by **non-authorized people** and, more importantly, prevent a **malicious hacker** from gaining access to the system and sending old views to the homeowner while a thief is breaking in. The areas of control are far more critical for enterprise-sensitive applications such as **healthcare monitoring** of patients and **banking applications**.



[How Security Cameras Can Help With Hospital Security and Patient Care Monitoring - Vulcan Security Systems](#)



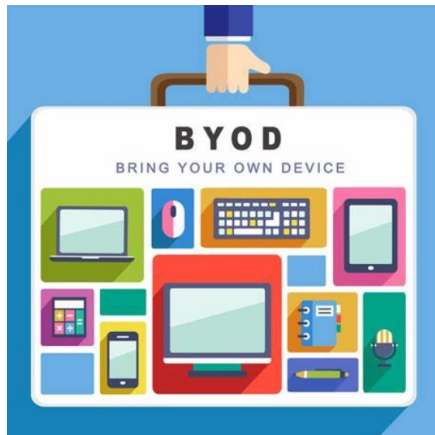
1.1. Tổng quan về công nghệ Internet vạn vật

How Is Security Guaranteed?

Securing IoT is perhaps the **biggest opportunity** for **technology companies** and will remain so far some time in the future. **Before IoT**, **information technology** security professionals worked in a bubble as they literally owned and controlled their entire networks and secured all devices behind firewalls.

With IoT, data will be collected from **external**, often **mobile**, **sensors** that are placed in public sites (e.g., city streets) allowing **strangers to send harmful data to any network**.

Bring your own device (BYOD) is another example where **third-party devices** and hence noncorporate data sources are allowed to enter the network.



1.1. Tổng quan về công nghệ Internet vạn vật

How Is Security Guaranteed?

IoT areas that are considered to be most vulnerable include:

- ❑ Accessing data during transport (network and transport security). Data will be transported in IoT networks at all time, for example, [from sensors to gateways](#) and [from gateways to data centers in enterprises](#) or [from sensors to gateways](#) for residential services such as video from home monitoring system to the homeowner's smartphone while he is in a coffee shop. This data may be sniffed by the man in the middle unless the transport protocols are fully secured and encrypted.
- ❑ Having control of IoT devices (control of the APIs) allows unauthorized persons to take full control of entire networks. Examples include shutting down cameras at home and shutting down patient monitoring systems

1.1. Tổng quan về công nghệ Internet vạn vật

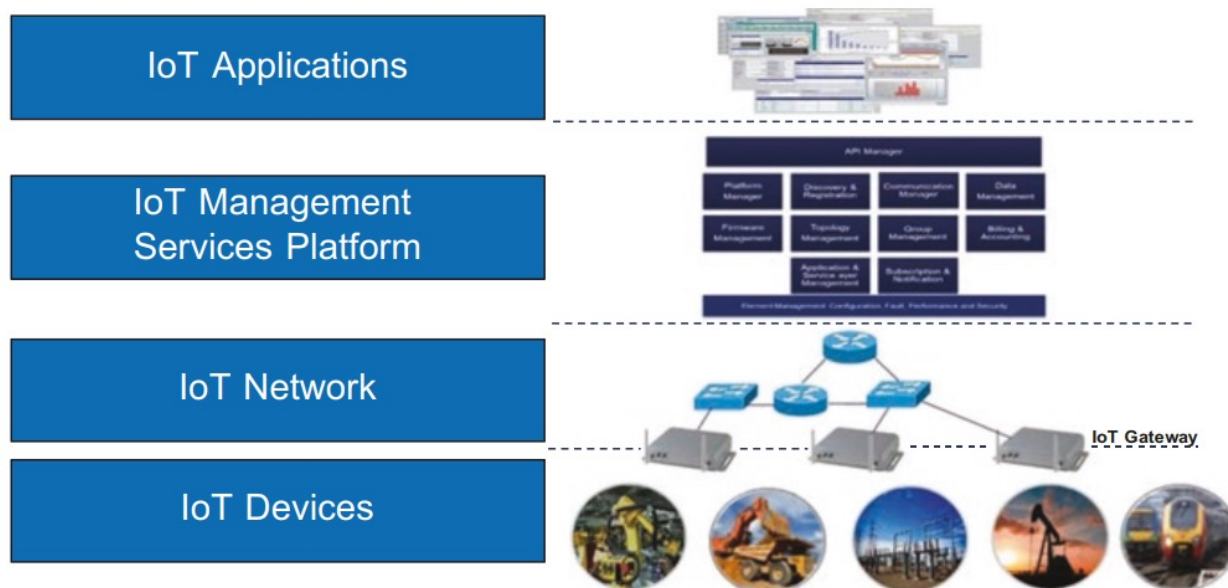
How Is Security Guaranteed?

IoT areas that are considered to be most vulnerable include:

- ❑ Having access to the IoT data itself. Is the data easily accessible? Is it stored encrypted?
Shared storage in the cloud is another problem where customer may log in as customer B and look at his data. Another common problem is spoofing data via Bluetooth. Many companies are adding Bluetooth support to their devices making it more feasible for unauthorized persons to access the device's data.
- ❑ Stealing official user or network identity (stealing user or network credentials). Many websites provide default passwords for vendors.

1.1. Tổng quan về công nghệ Internet vạn vật

1. ***IoT Device Level*** includes all IoT sensors and actuators (i.e., the Things in IoT).
2. ***IoT Network Level*** includes all IoT network components including IoT gateways, routers, switches, etc.
3. ***IoT Application Services Platform Level*** The functions of the IoT Services Platform include the ability to *deploy, configure, troubleshoot, secure, manage, and monitor IoT devices*.
4. ***IoT Application Level*** includes all applications operating in the IoT network.



1.1. Tổng quan về công nghệ Internet vạn vật

Advantages of the proposed IoT four-level model include:

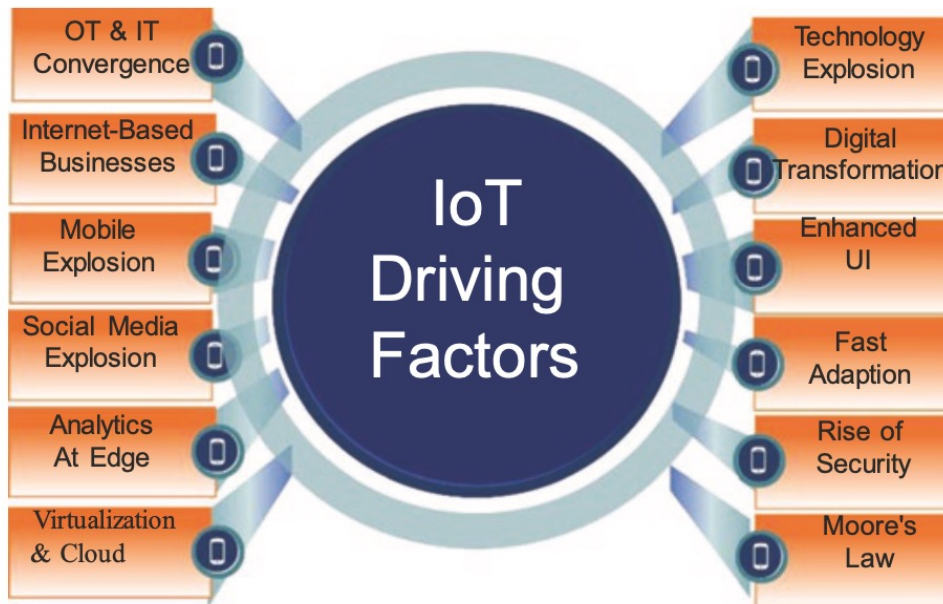
- ❖ Reduced Complexity: It breaks IoT elements and communication processes into smaller and simpler components, thereby helping IoT component development, design, and troubleshooting.
- ❖ Standardized Components and Interfaces: The model standardizes the specific components within each level (e.g., what are the key components for general IoT Services Platform) as well as the interfaces between the various levels. This would allow different vendors to develop joint solutions and common support models.
- ❖ Module Engineering: It allows various types of IoT hardware and software systems to communicate with each other.

1.1. Tổng quan về công nghệ Internet vạn vật

- ❖ Interoperability between vendors by ensuring the various technology building blocks can interwork and interoperate.
- ❖ Accelerate Innovation: It allows developers to focus on solving the main problem at hand without worrying about basic functions that can be implemented once across different business verticals.
- ❖ Simplified Education: It breaks down the overall complex IoT solution into smaller more manageable components to make learning easier.

1.1. Tổng quan về công nghệ Internet vạn vật

- ❖ IoT has already become a powerful force for business transformation, and its disruptive impact is already felt across all industries and all areas of society.
- ❖ There is a perfect storm of market disruptions happening at an unprecedented pace triggered by technology as well as new business and social requirements.

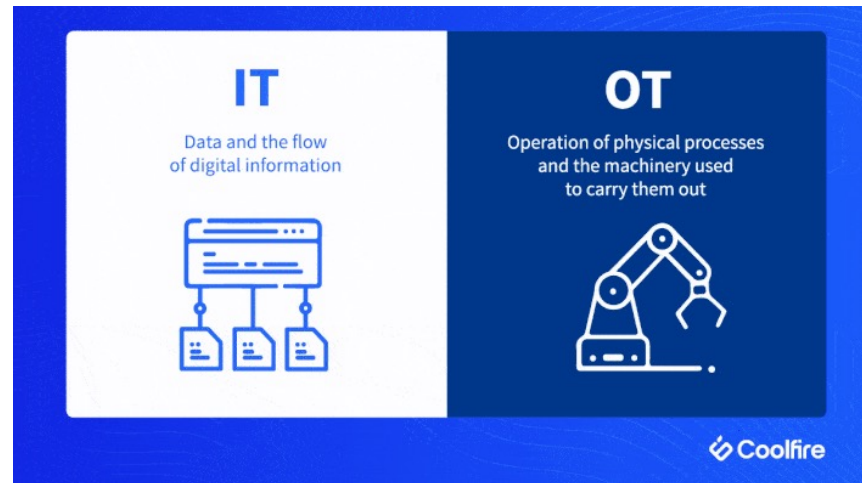


IoT 12 driving factors

1.1. Tổng quan về công nghệ Internet vạn vật

Convergence of IT and OT

- ❖ Operation technology (OT) is the world of industrial plants and industrial control and automation equipment that include machines and systems to run the *business*, *controllers*, *sensors*, and *actuators*.
- ❖ Information technology (IT) is the world of end- to-end information systems focusing on compute, data storage, and networking to support business operation in some context such as *business process automation systems*, *customer relation management (CRM) systems*, *supply chain management systems*, *logistics systems*, and *human resources systems*.



1.1. Tổng quan về công nghệ Internet vạn vật

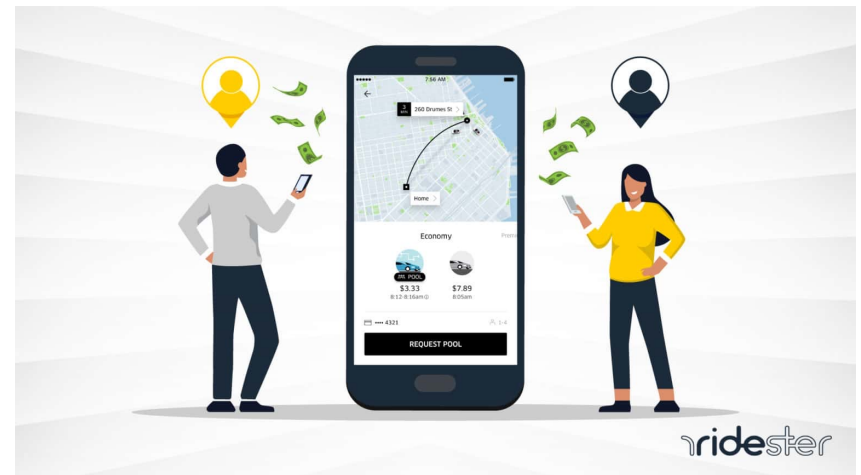
Convergence of IT and OT

- ❖ Historically, IT and OT were always managed by two separate organizations with different cultures, philosophies, and set of technologies.
- ❖ IT departments were originally created by companies to create efficient and effective forms of telephony communication among various departments. Then they were extended to provide video and web conferences and network internal communications and secure external electronic communications such as emails. Often the final decision with the selection of communication systems, website hosting, and backup servers was the responsibility of the IT department.
- ❖ OT relies on real-time data that drives safety, security, and control. It depends on very well-defined, tested, and trusted processes. Many plants need to run 24×7 with zero downtime (e.g., City Water Filtration System), and thus industrial processes cannot tolerate shutdown for software updates. IT is more lenient with software updates, introduction of new technologies, etc.

1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

The key component of Uber's solution is the Internet-based platform connecting customers (passengers) with the service providers (car drivers). Because the consumers are not Uber's employees and because there is practically an infinite number of cars that could potentially join Uber, Uber has the requirement to scale at an incredibly fast rate at zero marginal cost.



1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Uber*

Uber uses sensor technologies in driver's smartphones to track their behaviors. If you ride with Uber and your driver speeds, breaks too hard, or takes you on a wildly lengthy route to your destination, it is no longer your word against theirs. Uber is using *Gyrometer* and GPS data to track the behavior of their drivers. *Gyrometers* in smartphones measure small movements, while GPS combined with accelerometers shows how often a vehicle starts and stops and the overall speed.

1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ Uber Eat



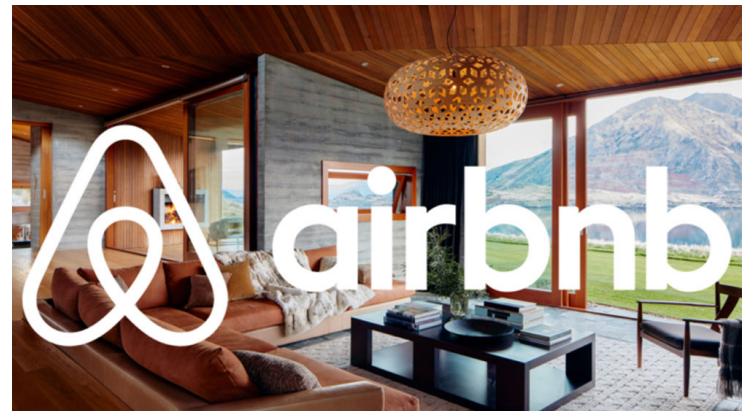
Today Uber is one of the leading transportation services in the world with a market value over **20 billion dollars**.

1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Airbnb*

Airbnb is an Internet-based service for people to list, find, and rent lodging. It was founded in **2008** in **San Francisco, California**, by **Brian Check** and **Joe Gebbia** shortly after creating *AirBed* and Breakfast during a conference. The original site offered *rooms*, *breakfast*, and *business networking opportunity* for the conference attendees who were unable to find a hotel. Shortly thereafter, the newly created company focused on high-profile events where alternative lodging was very limited.



1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Amazon*

Amazon.com is the largest Internet retailer company in the world started, in 1994, as an Internet-based book seller and swiftly expanded into music, movies, electronics, and household goods; Amazon utilized the Internet to break the traditional retailer model. It **did not need** to *stock many of the merchandises* it was selling on its website. Instead, it identified matching partner companies and issued customer orders over a secure Internet-based platform.



1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Amazon*



1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Tesla*

Tesla Motors was founded in 2003 by a group of engineers in Silicon Valley with a mission to develop a successful luxurious electrical car and then invest the resulting profits to make a less expensive electric car. With instant torque, incredible power, and zero emissions, Tesla's products would be cars without compromise.

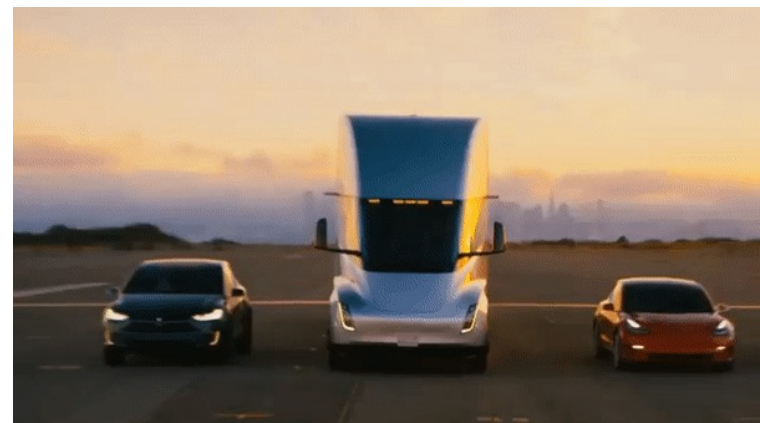


1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Tesla*

Tesla's engineers first designed a power train for a sports car built around an AC induction motor, patented in 1888 by **Nikola Tesla**, the inventor who inspired the company's name. The resulting Tesla Roadster was launched in 2008 with an incredible range of *245 miles per charge* of its *lithium ion battery*. The Roadster was able to set a new standard for electric mobility. In 2012, Tesla launched **Model S**, the world's first premium electric sedan.

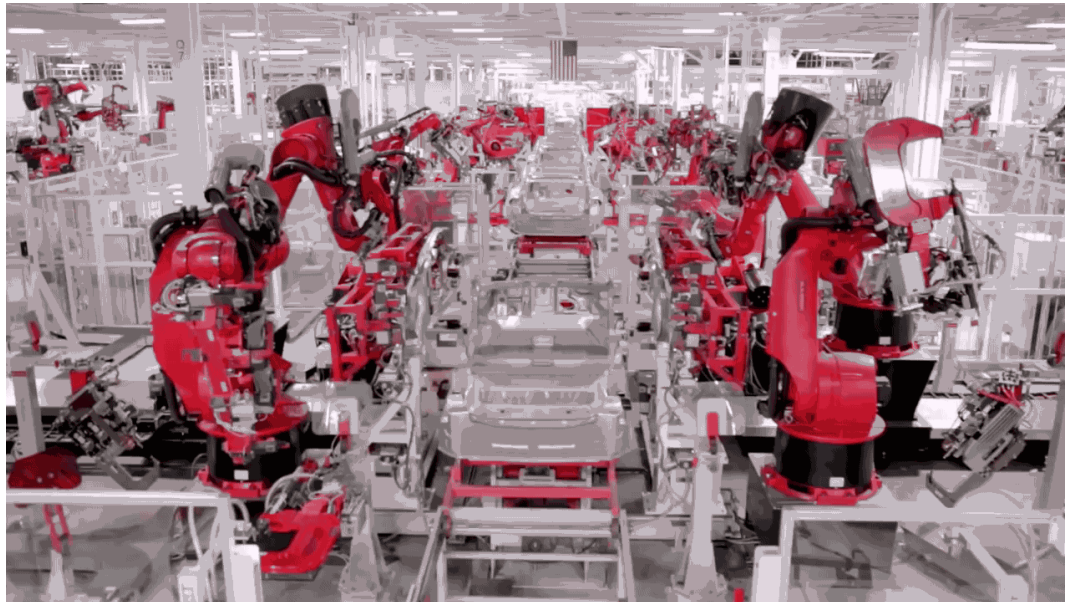


1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Tesla*

Tesla is considered as the best example yet of IoT. It did not only bend the traditional industry manufacturing model to Internet-based model with *thousands of sensors*.



1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Self-Driving Cars*

Self-driving cars can be divided into two main categories: semiautonomous and fully autonomous. A semiautonomous car performs certain self-driving tasks (e.g., fully brakes when it gets too close to an object, drives itself on the freeway), while a fully autonomous car drives itself from origin to destination without any driver interaction. Fully autonomous cars are further divided into user-operated and driverless.



1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Self-Driving Cars*

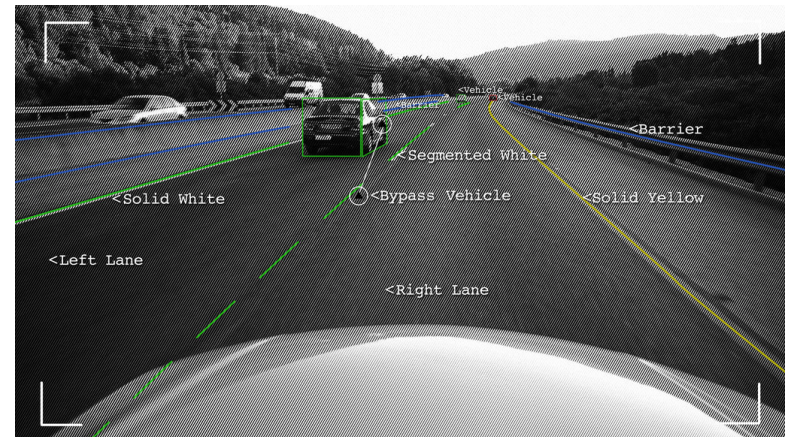
Safety is considered one of the biggest advantages of self-driving cars. In general, self-driving cars are equipped with a large number of sensors including laser range finders (to measure a subject's distance and take photos that are in sharp focus), radars, and video cameras collecting information from the road. They are also equipped with actuators to control steering and braking. The collected data (from sensors, radars, and video) is promptly processed with the positional information from the car's GPS unit and the navigation system to determine its position and to build a three-dimensional model of its surroundings.

1.1. Tổng quan về công nghệ Internet vạn vật

The Astonishing Introduction of Creative Internet-Based Businesses

❖ *Self-Driving Cars*

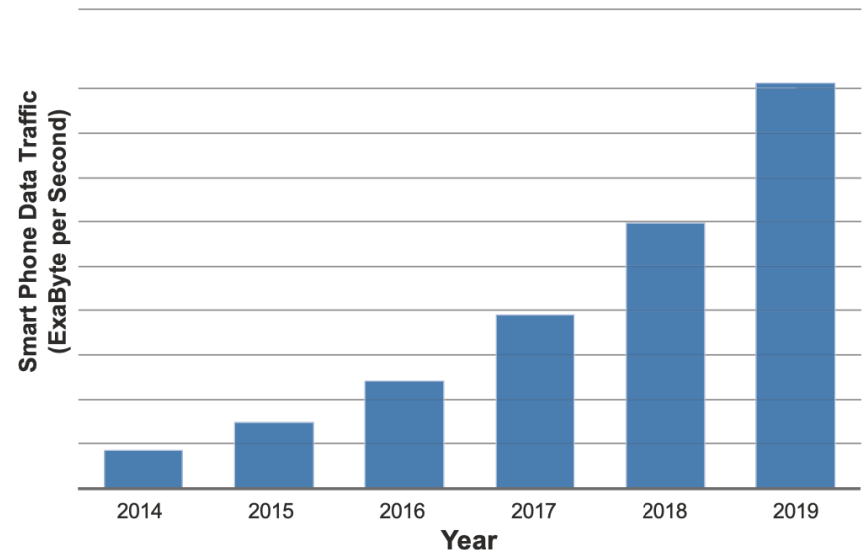
The resulting model is then processed by the car's control system to make navigation decisions. Self-driving car control systems typically use stored maps to find optimal path to destination, avoid obstacles, and send decisions to the car's actuators. IoT applies to interactions and communications between self-driving care components, between the *car* and *roadside infrastructure*, as well as among *self-driving cars*.



1.1. Tổng quan về công nghệ Internet vạn vật

Mobile Device Explosion

Mobile data traffic has grown 18-fold in the last few years. The increase in mobile data traffic is driven by two factors: the increase in the *number of users* and the *data consumption per user*. The average smartphone generated 4 GB of traffic per month in 2019. This growth is fueled by IoT connecting things with people and more importantly allowing people to monitor and control things from anywhere in the world in real time.



Smartphone traffic over time

1.1. Tổng quan về công nghệ Internet vạn vật

Social Network Explosion

Social networks, such as Facebook, Instagram, Twitter, and YouTube, and the adoption of cloud-based services, such as Amazon's AWS and Salesforce.com, are all examples of the large-scale migration to the cloud across virtually every industry. In fact, two-thirds of all data center traffic will be from the cloud in 3 years. All of this leads to data explosion, where, already, the data being created on the Internet each day is equal to half of all the data that has been accumulated since the dawn of humanity.



<https://aws.amazon.com/vi/what-is-aws/>

1.1. Tổng quan về công nghệ Internet vạn vật

Analytics at the Edge

Before introducing the different versions of analytics, it is important to define the terms: *big data*, *structured data*, and *unstructured data*. Big data refers to the extremely large amount of data being generated and accumulated by IT systems as the result of the operation of an associated system. The latter could be a product, process, service, etc. This massive amount of data can be analyzed to identify patterns and gain insights into the operation of the associated system. The analysis often involves applying statistical techniques since human processing is not viable due to the sheer volume of the data.



1.1. Tổng quan về công nghệ Internet vạn vật

Analytics at the Edge

Structured data refers to organized data that can fit in rows and columns. Examples of such data include customer data, sales data, and stock records. Structured data is often high value, cleansed, and indexed. Unstructured data, on the other hand, is difficult to organize or bring together. Examples of unstructured data include *images*, *X-rays*, *video*, *social media data*, and *some machine outputs mixed with text*.

STRUCTURED DATA



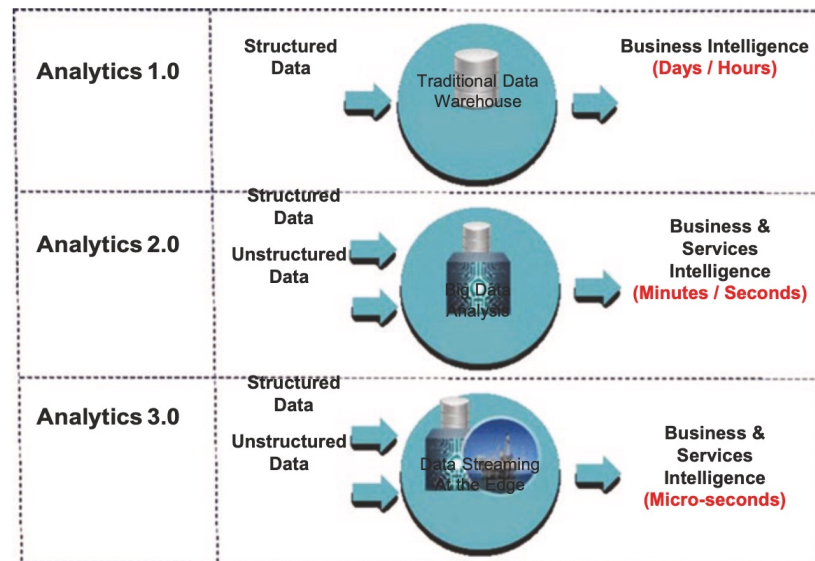
UNSTRUCTURED DATA



1.1. Tổng quan về công nghệ Internet vạn vật

Comparison of key factors for Analytics 1.0, 2.0, and 3.0

	Analytics 1.0	Analytics 2.0	Analytics 3.0
Collected data type	Structured	Structured and unstructured	Structured and unstructured
Data analysis location	Centralized data center	Centralized data center	At edge and in data center
Time to analyze data	Days–hours	Hours–minutes	Seconds–microseconds
Data volume	Small data	Big data	Big data



Analytics 1.0, 2.0, and 3.0

1.1. Tổng quan về công nghệ Internet vạn vật

Cloud Computing and Virtualization

Cloud computing, which was introduced in 2008, allows enterprises to outsource their computing infrastructure fully or partially to public cloud providers (e.g., Amazon AWS, Microsoft Azure, Google Compute Engine).

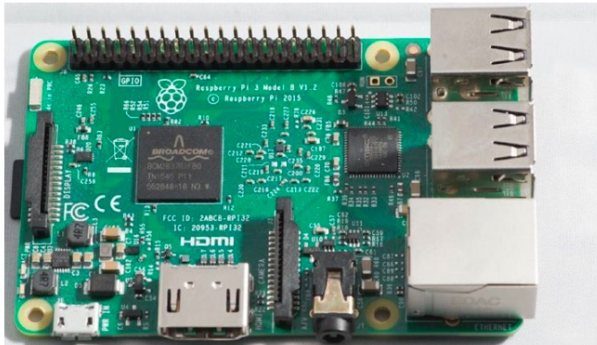


**Google
Compute
Engine**

1.1. Tổng quan về công nghệ Internet vạn vật

Technology Explosion

IoT hardware (e.g., sensors, inexpensive computers such as Raspberry Pi, open - source microcontrollers such as Arduino) and software technologies are not only being developed faster than ever before but with much lower prices. Such devices are already transforming user behaviors and creating new business opportunities.



Raspberry Pi B+



Raspberry Pi Zero



Arduino

1.1. Tổng quan về công nghệ Internet vạn vật

Digital Convergence/Transformation

Digital convergence has initially started with a *limited scope*: move to “*paperless*” operation and save trees.

Convergence is a concept that describes how previously separate *technologies, processes, and data* combine to create new products, services, end experiences that reshape industries.

The smartphone is a classic example of convergence. It's a camera, address book, map, and media player.

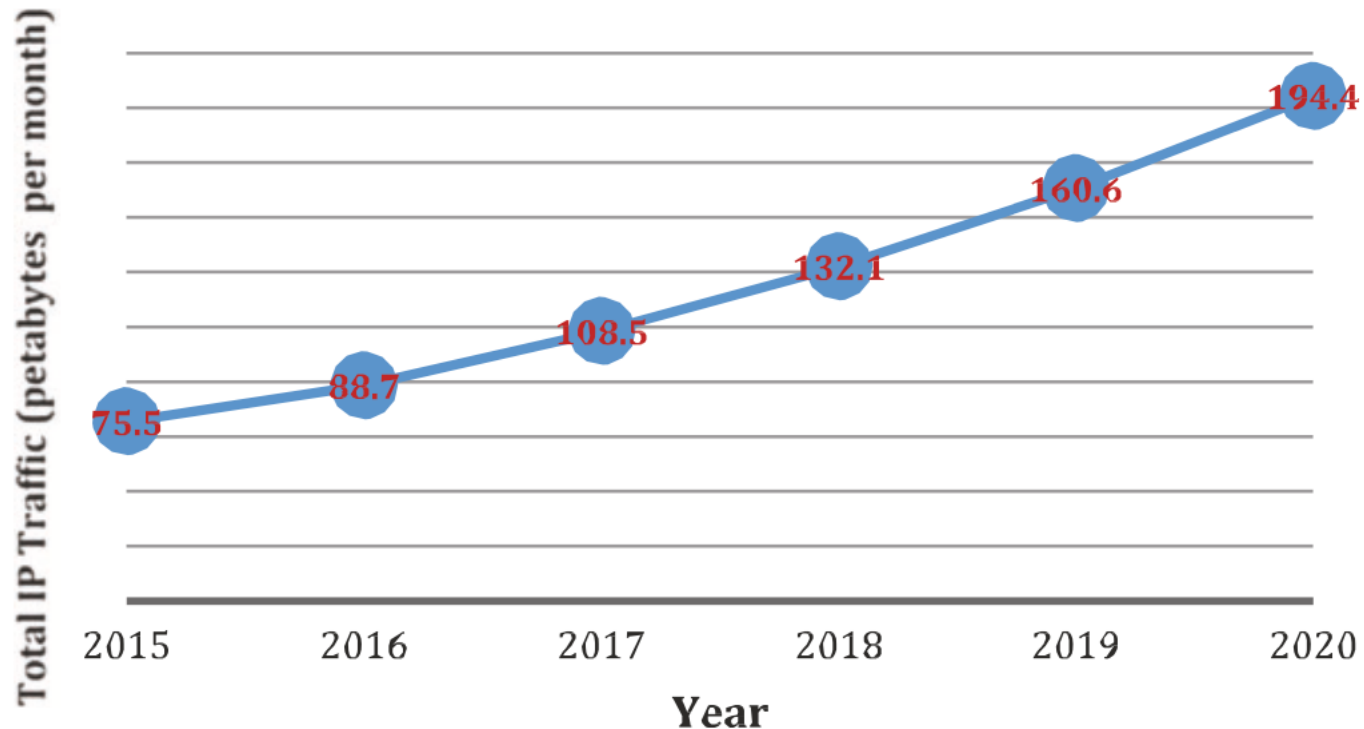
1.1. Tổng quan về công nghệ Internet vạn vật

Enhanced User Interfaces



1.1. Tổng quan về công nghệ Internet vạn vật

Fast Rate of IoT Technology Adoption (Five Times More than Electricity and Telephony)



Global IP traffic growth, 2015–2020.

1.1. Tổng quan về công nghệ Internet vạn vật

The Rise of Security Requirements

Protection of business and personal data and systems has been an issue since the inception of data networks. With the commercialization of the Internet, security concerns expanded to cover personal privacy, financial transactions, and the threat of cyber robbery. Today, security of the network is being expanded to include safety or physical security.

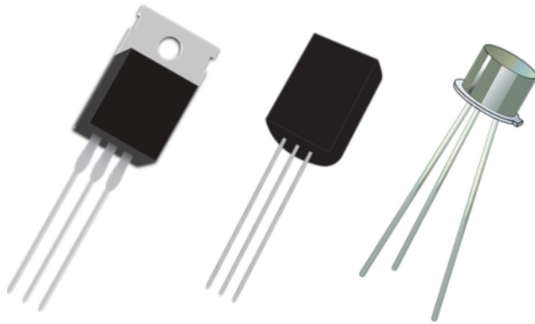


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The Nonstop Moore's Law

It is possible to summarize *Moore's Law* impact with *three key observation*:

1. Over the history of computing hardware, computer power has been doubling approximately every 18 months.
2. Silicon transistor storage technology size has continued to shrink over the years and is approaching atomic level. For years now, we have been putting more power and more storage on the same size device.
3. The price of the transistor is being reduced by more than 50% every year.



https://vi.wikipedia.org/wiki/Định_luật_Moore

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The Nonstop Moore's Law



1.1. Tổng quan về công nghệ Internet vạn vật

Transport

- Reducing travel times by analyzing traffic
- Ride-sharing apps-determining the price, supply chain prediction
- Autonomous vehicles



Artificial Intelligence & Autopilot | Tesla

Workplace

- Robotics in manufacturing
- Automated safety checks in factories
- Autonomous haulage
- Enriched recruitment
- Automated timesheets(e.g., Blackbelt)



Pepper the humanoid and programmable robot | SoftBank Robotics

1.1. Tổng quan về công nghệ Internet vạn vật

Education

- Plagiarism checkers
- Automated grading
- Customized digital learning interfaces
- Virtual teachers or lecturers



Sport

- Wearable tech to analyze performance
- Smart ticketing
- Automated video highlights
- Computer vision referee

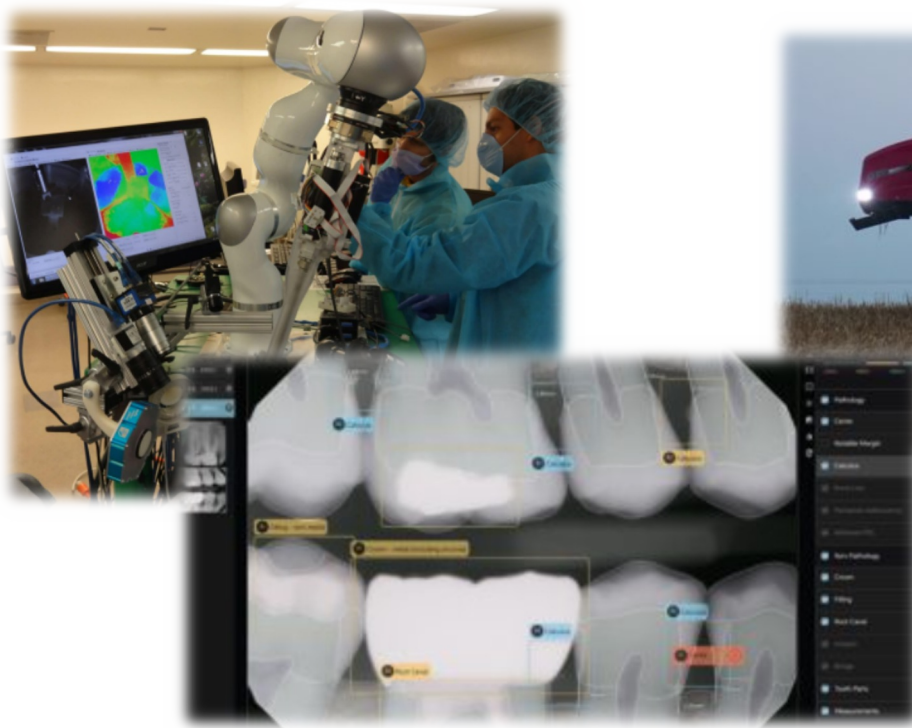


Five Impacts of AI on the Education Sector - Wavel

1.1. Tổng quan về công nghệ Internet vạn vật

Healthcare

- Autonomous surgical robots
- Automatic disease identification and diagnosis
- Epidemic outbreak prediction



Agriculture

- Robot harvesters
- Computer vision to monitor crop and soil health
- Predictive analysis for environmental impacts on crops



[Greeneye Helps Farmers Reduce Chemicals and Costs | NVIDIA Blog](#)

[- Archives Page 1 | NVIDIA Blog](#)

1.1. Tổng quan về công nghệ Internet vạn vật

Entertainment

- Music suggestions (Spotify, Apple Music, Google Play music)
- Automatic music creation
- Film and TV suggestions (Netflix, Amazon Prime, Hulu)



Spotify – Web Player

Smart homes

- Personal assistants
- Automatic goods ordering
- Home security
- Temperature and light control



Bixby | Apps - The Official Samsung Galaxy Site

1.1. Tổng quan về công nghệ Internet vạn vật

Defence

- Unmanned Aerial Vehicles (UAVs)
- Civilian detection
- Autonomous decision making
- Target identification
- Diagnosis and maintenance of weapons systems



AI in Military Drones and UAVs – Current Applications
| Emerj Artificial Intelligence Research

Army Buys 9,000 Mini-Drones, Rethinks Ground Robots - Breaking Defense



Russia's Autonomous Robot Tank Passes New Milestone (And Launches Drone Swarm) (forbes.com)



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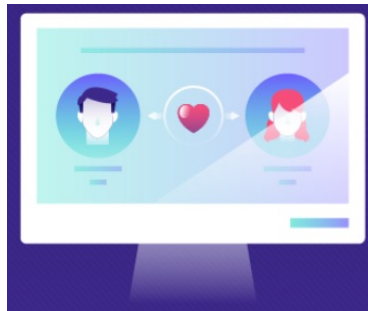
Automatic transportation

The goal is for all transportation to become automated in the future



Solving climate change

Using big data, AI and IoT could one day identify trends and use that information to present solutions to the world's biggest problems like climate change



Cyborg technology

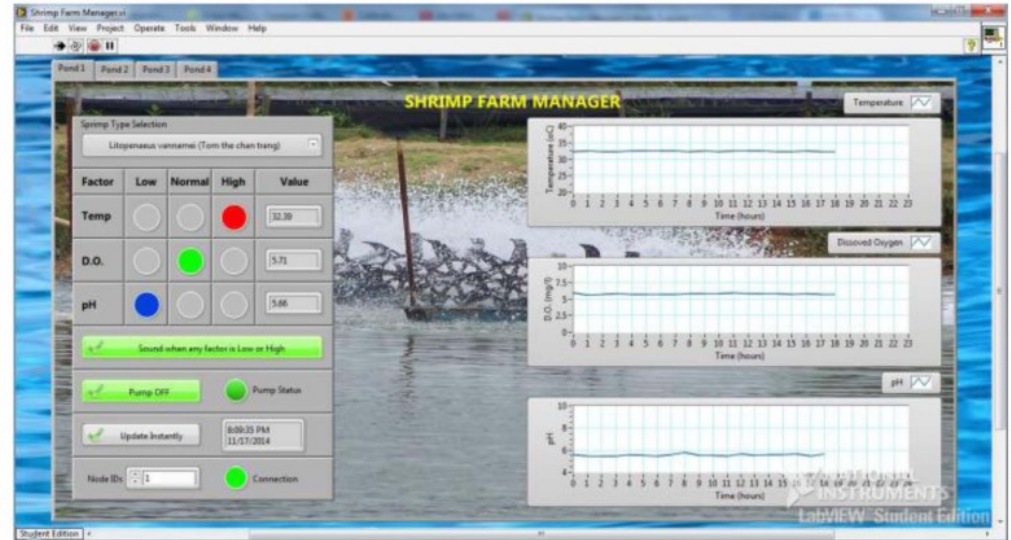
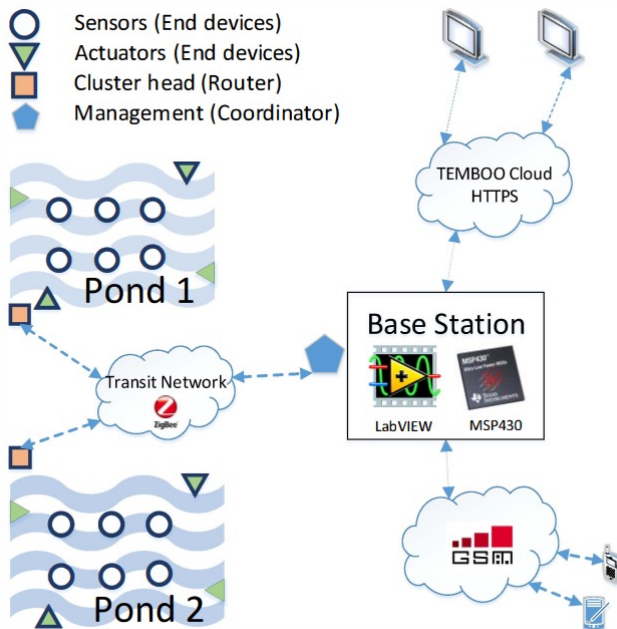
AI and robotics will help us overcome our cognitive and physical limitations, including robotic limbs capable of communicating with our brains



Predicting the future

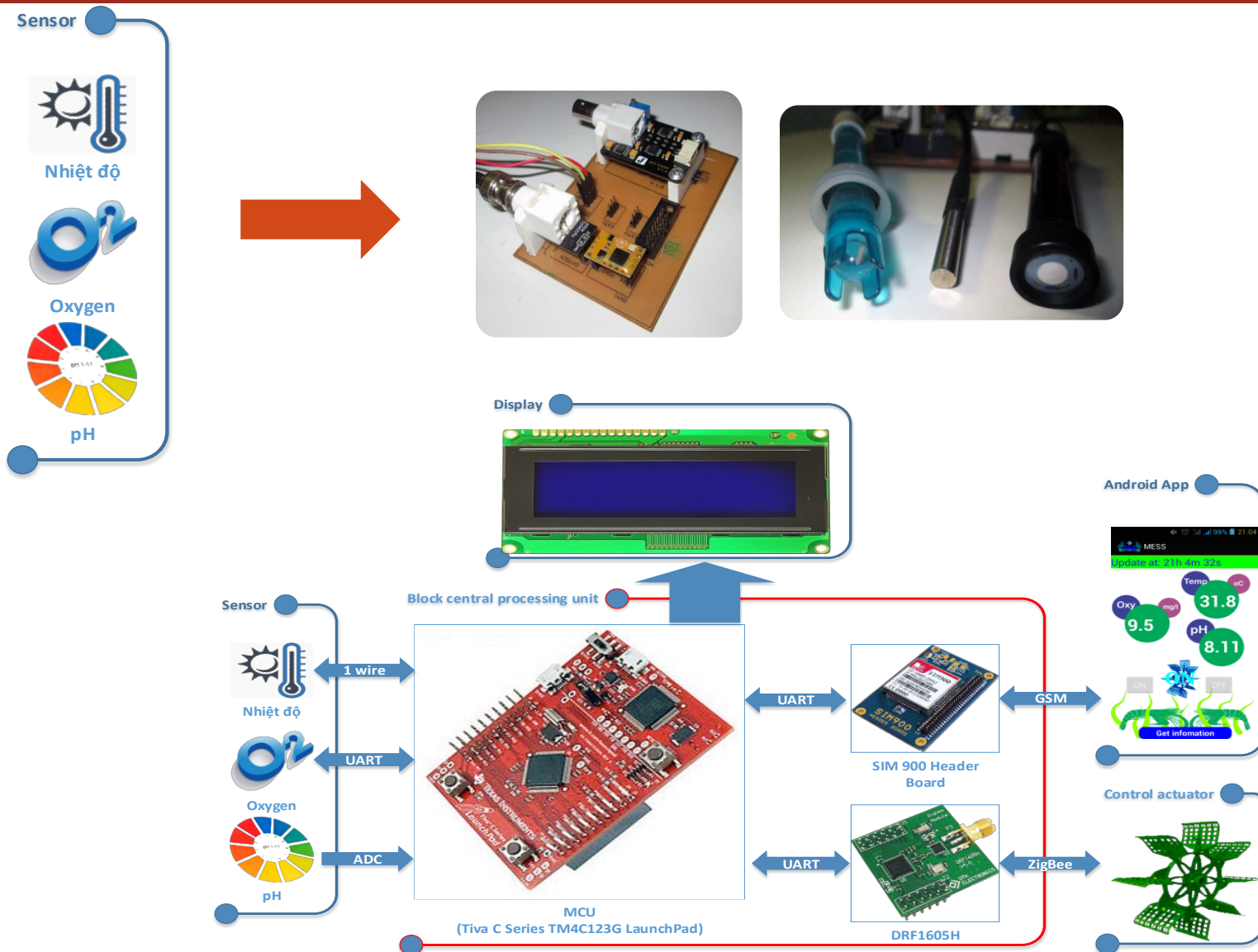
Machine learning uses the past to make predictions about the future. It will be able to predict things as intimate as who will start dating or who will get divorced.

1.1. Tổng quan về công nghệ Internet vạn vật

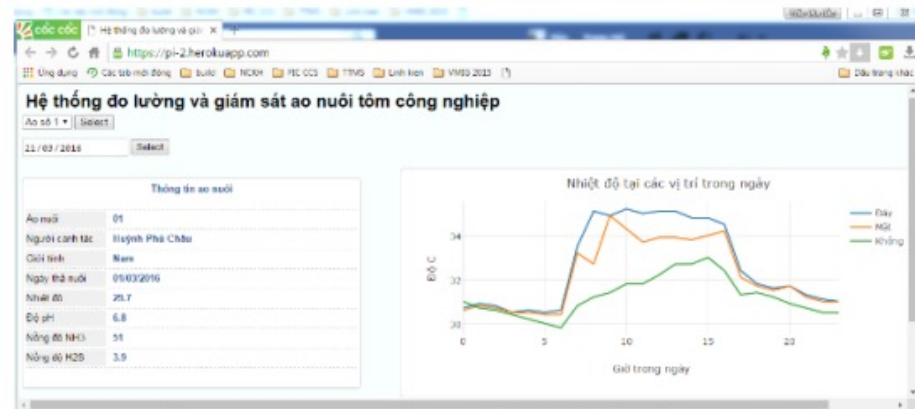
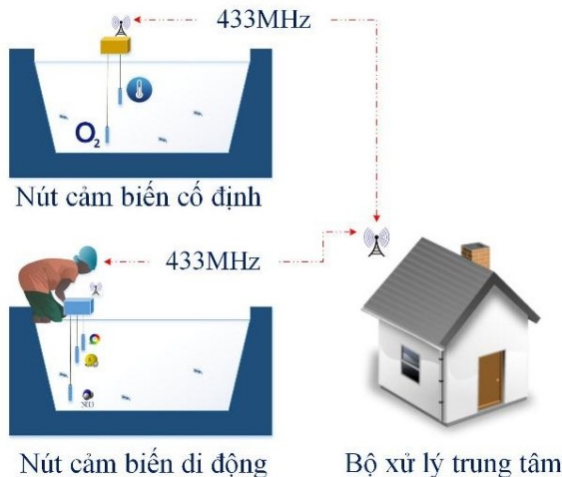
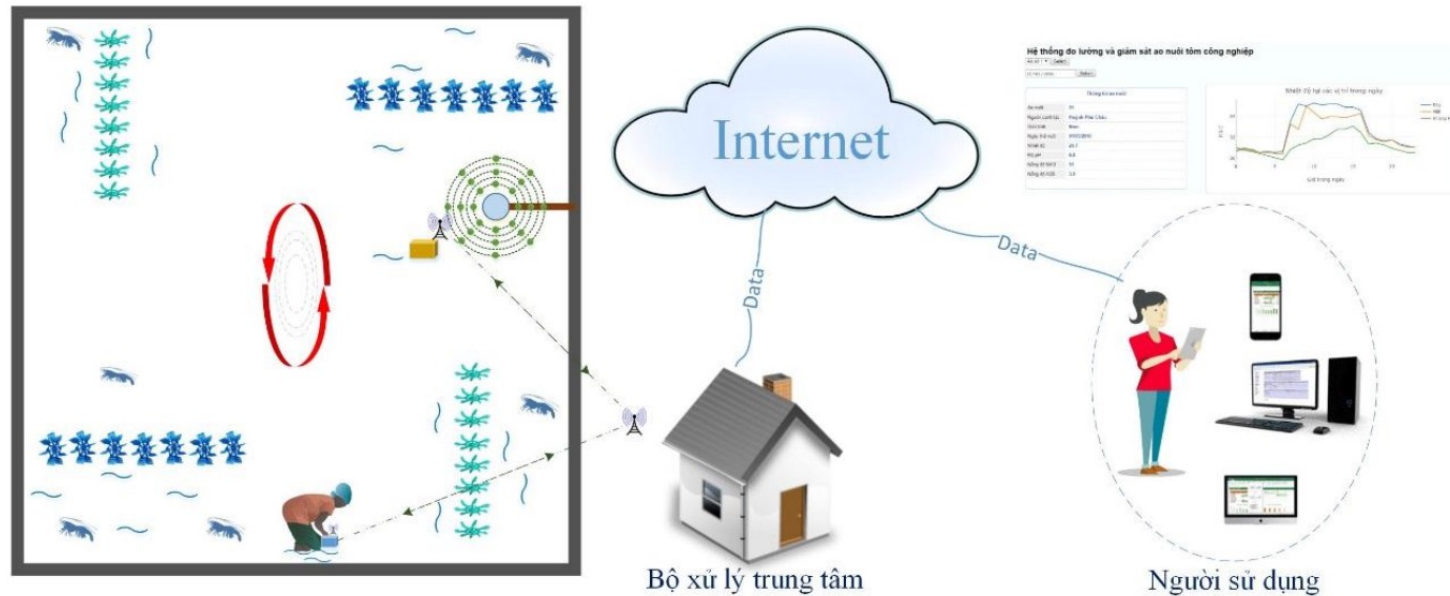


This system is capable of collecting, analyzing and presenting data on a **Graphical User Interface (GUI)**, programmed with **LabVIEW**. It also allows the user to get the updated **sensor information online** based on **Google Spreadsheets** application, via Internet connectivity, or at any time through the **SMS gateway** service and sends alert message promptly enabling user interventions when needed.

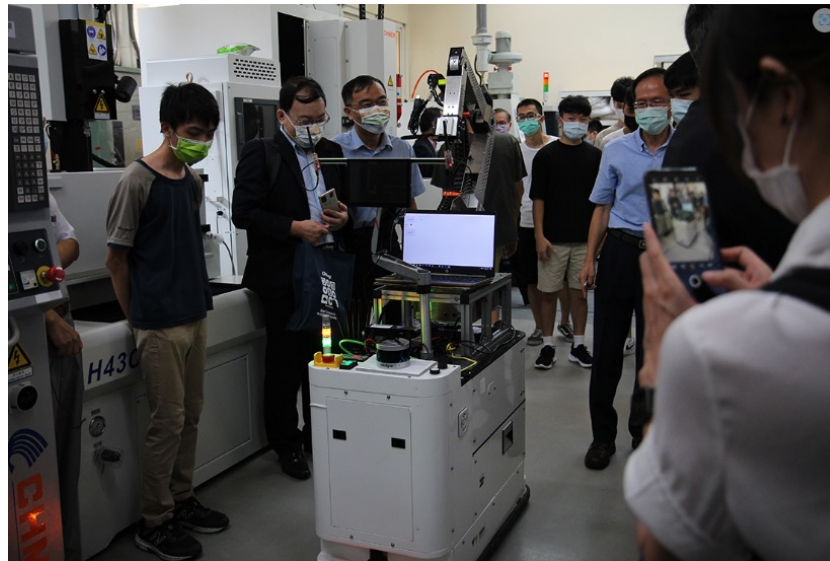
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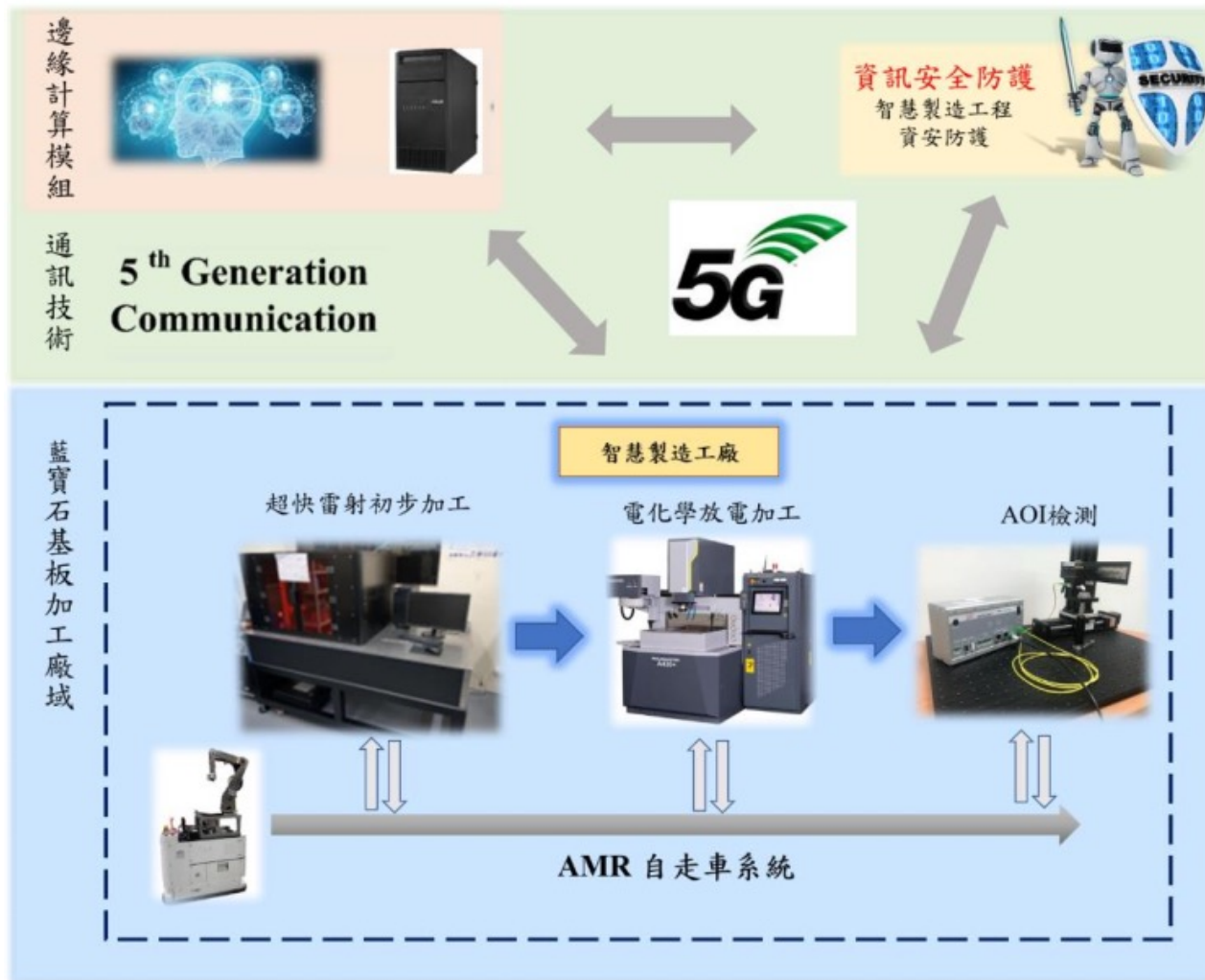


ORing, NCU (National Central University), and Chunghwa Telecom Join Hands on 5G Smart Manufacturing

[ORing / News / Press Release \(oringnet.com\)](http://oringnet.com)

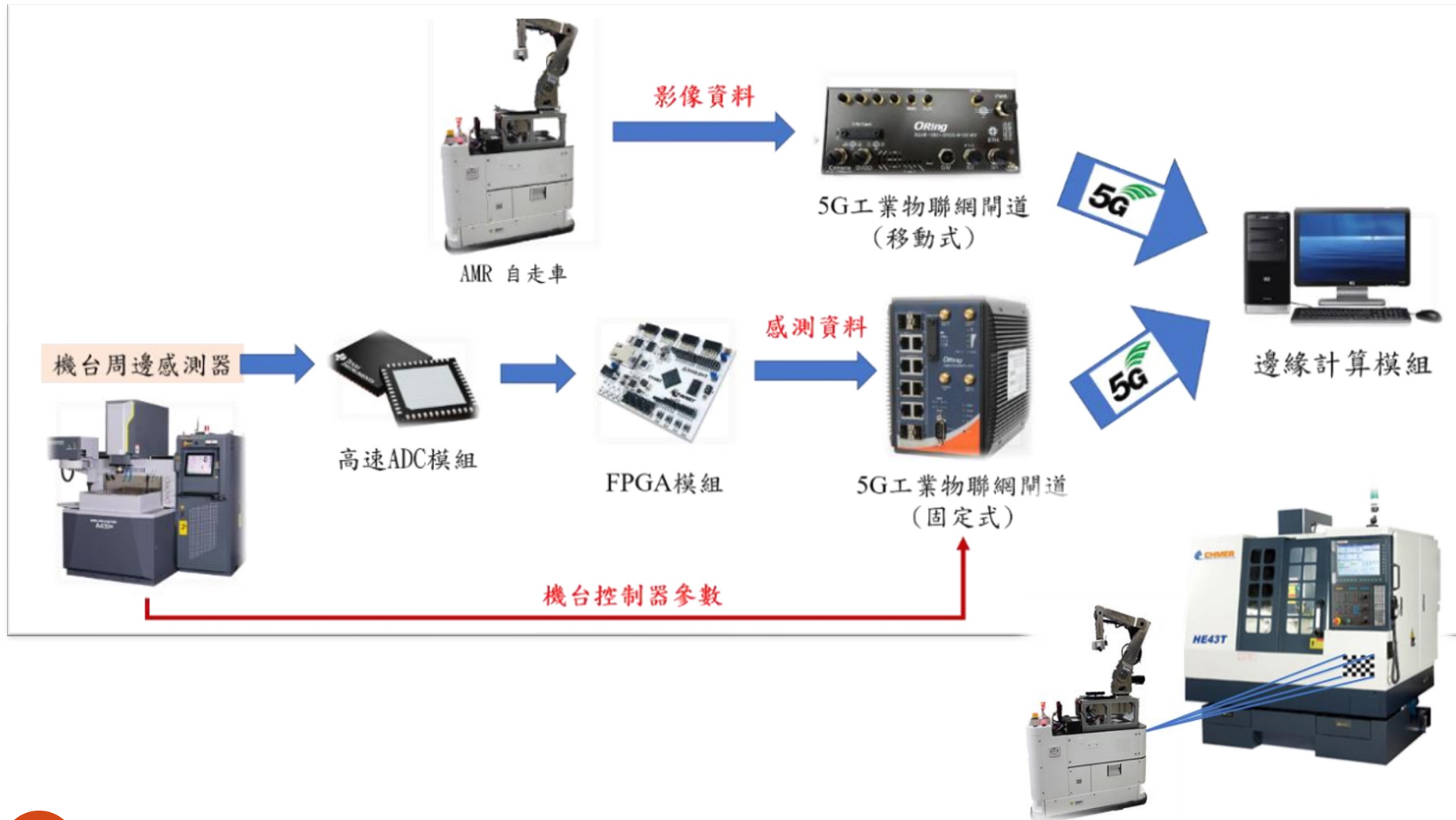
The **Department of Mechanical Engineering** has established a smart factory based on **Industry 4.0** under the support of Jing-Yang Jou, President of NCU, to develop **laser, discharge, and electrochemical machining technologies**. The **smart factory** has integrated a variety of smart technologies including **AI, big data analytics, and edge computing**. [Edge computing là gì? Mô hình kiến trúc trong Edge computing \(bizflycloud.vn\)](http://bizflycloud.vn)

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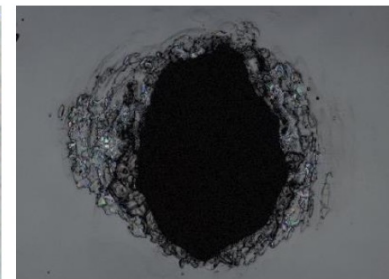
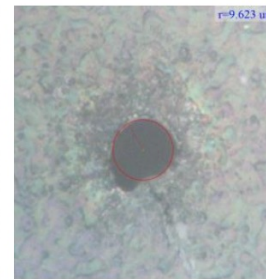
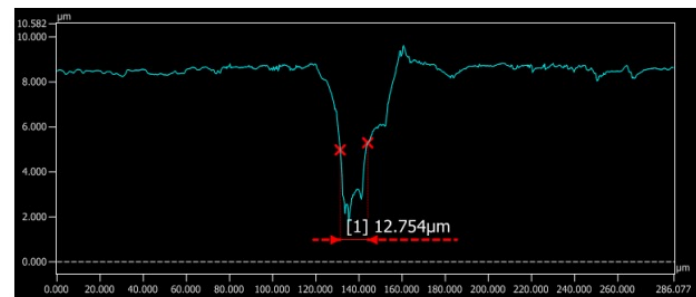
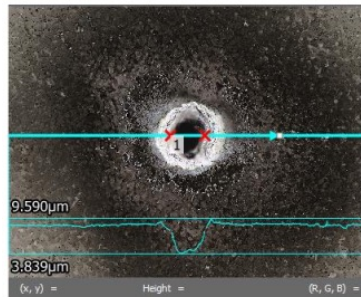
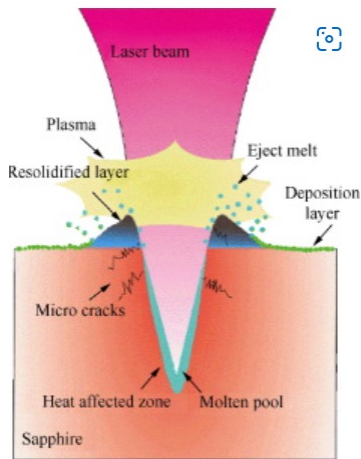
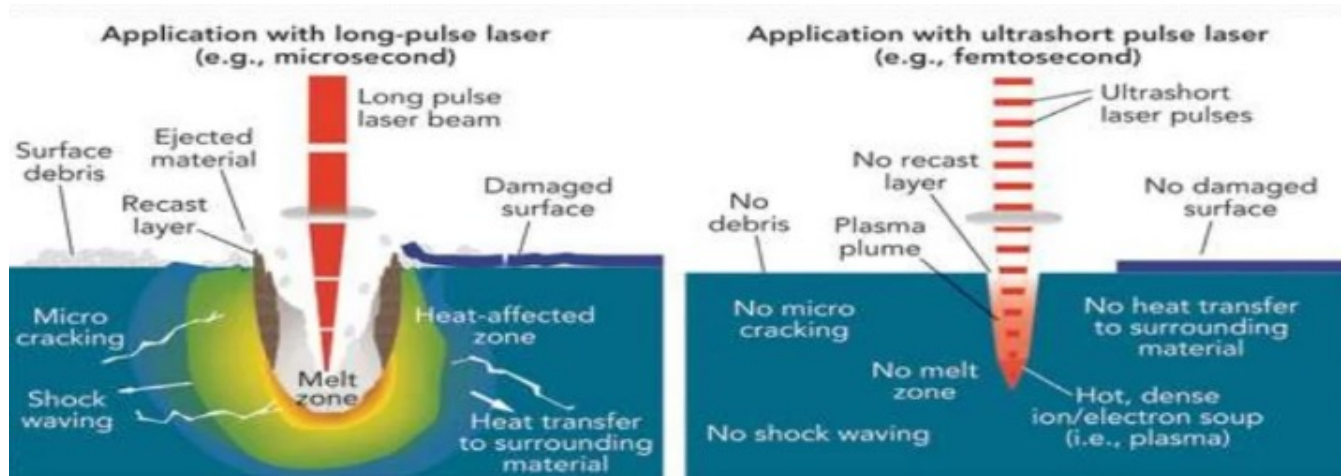


Automated
optical
inspection

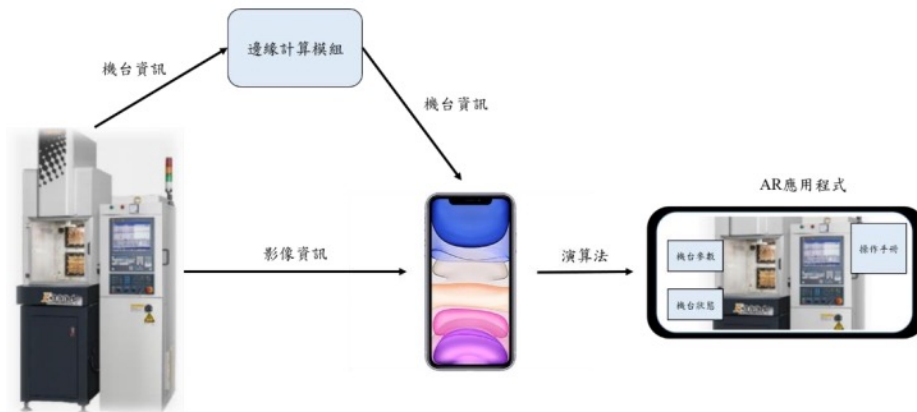
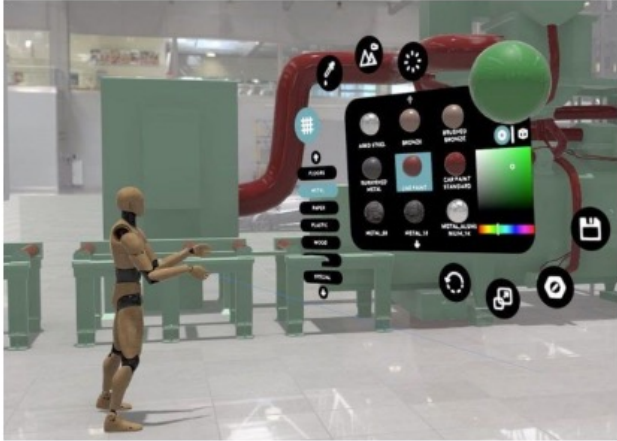
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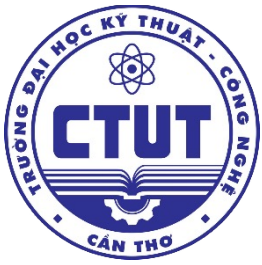


1.2. Xu thế ứng dụng IoTs và các công nghệ liên quan cho đô thị thông minh ở Việt Nam

- Công nghệ IoTs/5G và kỹ thuật xử lý dữ liệu lớn (Big Data), trí tuệ nhân tạo (AI) đóng vai trò quan trọng trong việc tạo ra những đổi mới đột phá trong CNTT và truyền thông, và các lĩnh vực ứng dụng liên quan.
- Thành phố thông minh là một trong những ứng dụng như vậy, đã thu hút được sự chú ý lớn từ các chính phủ, cộng đồng ngành học thuật và CNTT tại nhiều quốc gia. Đặc biệt, Big Data/AI và IoT/5G cung cấp các công cụ và công nghệ cho phép quan trọng để thúc đẩy sự chuyển đổi của các thành phố lớn trên toàn thế giới thành những nơi đáng sống, làm việc và tận hưởng.

1.2. Xu thế ứng dụng IoTs và các công nghệ liên quan cho đô thị thông minh ở Việt Nam

- Các công nghệ IoTs/5 mới, một khi được tích hợp đúng vào cơ sở hạ tầng thành phố thông minh (*Smart Cities*), có thể hỗ trợ số lượng lớn kết nối không dây từ các loại thiết bị không dây khác nhau (ví dụ: cảm biến, máy ảnh, thiết bị AR/VR, xe thông minh) thông qua môi trường mạng có tốc độ cao, truyền dữ liệu cực kỳ đáng tin cậy và độ trễ thấp.
- Tại VN, mặc dù còn nhiều bất cập, song hiện nay, các đô thị ở VN đang phát triển theo hướng tiếp cận với nhiều mô hình tiên tiến theo xu thế hội nhập quốc tế, tạo ra nhiều tiền đề về ứng dụng và phát triển CNTT, truyền thông để phát triển đô thị thông minh và quản lý thông minh.



***FACULTY OF MECHANICAL ENGINEERING
CAN THO UNIVERSITY OF TECHNOLOGY***



Thanks for your listening!

